Haofei Liang

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Education

Sept. 2022 – Dec. 2026 (expected) **Shanghai Jiao Tong University**
Eng.D. in Computer Science and Technology.
Research area: applications and optimization of fully homomorphic encryption schemes.
Advisor: Prof. Yu Yu

Sept. 2020 – June 2022 **Shanghai Jiao Tong University**
M.Sc. in Computer Science and Technology.
Advisor: Prof. Yu Yu

July 2019 – Aug. 2020 Preparation for the national graduate entrance examination.

Sept. 2015 – June 2019 **Wuhan University**
B.Sc. in Information Security.

Research Interests

My research focuses on **Fully Homomorphic Encryption (FHE)**, with particular emphasis on the **TFHE** family of schemes. Specific topics include:

Functional Bootstrapping Techniques to evaluate arbitrary functions during bootstrapping while keeping latency low.

Circuit Bootstrapping Efficient conversion between FHE ciphertext formats to support deeper homomorphic circuits.

Applications of FHE Privacy-preserving protocols built on FHE, including oblivious message retrieval and single secret leader election.

Publications

Conference Proceedings

- **Haofei Liang**, Zeyu Liu, Eran Tromer, Xiang Xie, Yu Yu. *InstantOMR: Oblivious Message Retrieval with Low Latency and Optimal Parallelizability*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. USENIX · ePrint: 2025/2317.
- Zhenkai Hu, **Haofei Liang**, Xiao Wang, Xiang Xie, Kang Yang, Yu Yu, Wenhao Zhang. *Ajax: Fast Threshold Fully Homomorphic Encryption without Noise Flooding*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. USENIX · ePrint: 2025/1834.
- Fengrun Liu, **Haofei Liang**, Xiang Xie, Yu Yu, Wenting Zheng, Yuncong Hu. *HasteBoots: Proving TFHE Programmable Bootstrapping in Seconds*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. USENIX · ePrint: 2025/261.
- **Haofei Liang**, Zeyu Liu, Yunhao Wang, Xiang Xie, Yu Yu, Fan Zhang. *Relect: Single Secret Leader Election via FHE with Reduced Computation and Communication and Transparent Setup*. In: 33rd ACM Conference on Computer and Communications Security (ACM CCS 2026), The Hague, The Netherlands, November 15–19, 2026. (Accepted, to appear) ePrint: 2026/1619.

Conference Talks

Aug. 2026 **USENIX Security 2026**, Baltimore, MD, USA – presented *InstantOMR*.

Nov. 2026 **ACM CCS 2026**, The Hague, The Netherlands – upcoming conference talk on *Relect*.

Collaborations

Prof. Fan Zhang, Zeyu Liu, and Yunhao Wang (Yale University); Prof. Eran Tromer (Boston University); Prof. Xiao Wang (Northwestern University); Fengrun Liu (Carnegie Mellon University); and Prof. Xiang Xie (East China Normal University).

Open-Source Projects

- **primus** – High-performance fully homomorphic encryption library currently under active development; lead developer.
- **tfhe-omr** – Reference implementation of the InstantOMR oblivious message retrieval scheme, supporting low latency and high parallelism; lead developer and first author of the corresponding paper.
- **Ajax** – Reference implementation of the Ajax threshold fully homomorphic encryption scheme without noise flooding; primary developer.
- **primus-fhe** – Open-source NTT-TFHE library; lead developer.
- **HasteBoots** – Open-source implementation of high-performance zero-knowledge proofs for TFHE programmable bootstrapping; primary developer. ePrint: 2025/261.
- **Relect** – Open-source implementation of a high-performance single secret leader election scheme without relying on heavy FHE schemes such as BFV or TFHE; lead developer and first author of the corresponding paper. ePrint: 2026/1619.

Skills

Programming Rust (primary; 3+ years of experience in FHE library and protocol development), C (small-scale projects), Python (data analysis, benchmarking scripts, and figure generation), LaTeX (academic writing).
Cryptographic Expertise Deep understanding of the TFHE scheme family (functional bootstrapping, circuit bootstrapping), noise analysis, and parameter tuning; working knowledge of BFV, BGV, and CKKS schemes.
FHE Tools & Libraries Primary development on an in-house Rust-based FHE library.
Languages Native Mandarin Chinese; English (CET-6; proficient reading and writing, conversational speaking).

Conferences Attended

2023 **ASIACRYPT 2023**, Guangzhou, China – attendee.
2023 **CHINACRYPT 2023**, Guangzhou, China – attendee.

Online Resources

- **Website:** haofeiliang.github.io
- **InstantOMR paper:** usenix.org/conference/usenixsecurity26/presentation/liang
- **InstantOMR ePrint:** eprint.iacr.org/2025/2317
- **InstantOMR code:** github.com/xiangxiecrypto/tfhe-omr
- **Ajax paper:** usenix.org/conference/usenixsecurity26/presentation/hu-zhenkai
- **Ajax ePrint:** eprint.iacr.org/2025/1834
- **Ajax code:** github.com/primus-labs/Ajax
- **HasteBoots paper:** usenix.org/conference/usenixsecurity26/presentation/liu-fengrun
- **HasteBoots ePrint:** eprint.iacr.org/2025/261
- **HasteBoots code:** github.com/f7ed/HasteBoots
- **Relect ePrint:** eprint.iacr.org/2026/1619
- **Relect code:** github.com/haofeiliang/new-ssle
- **primus code:** github.com/haofeiliang/primus
- **primus-fhe code:** github.com/primus-labs/primus-fhe